

TAVISHI GUPTA

Austin, Texas | tavishigupta888@gmail.com

EDUCATION

University of Texas at Austin — B.S. Electrical and Computer Engineering, Honors

Class of 2030

Round Rock High School

Class of 2026

GPA: 5.81 weighted / 4.0 unweighted | Rank: 27/986

EXPERIENCE

Chief Technology Officer, Femezi

May 2026 – Present

- Developed a full-stack JavaScript web application for e-commerce, enabling users to browse and purchase products through a responsive interface.
- Integrated MSG91 for phone number authentication (OTP), Razorpay for secure payment processing, and Shiprocket for automated shipping and order fulfillment.
- Deployed and maintained the application using Docker, Railway (backend), and Vercel (frontend), configuring production environments and deployment pipelines.
- Successfully completed 100+ orders, ensuring reliable authentication during payment and shipping.

Vice President / Team Lead, Robotics Club

August 2022 – May 2026

- Led the design, programming, and testing of competitive VEX V5 robots, earning qualification and placement at the State Championship.
- Developed autonomous navigation in C++ using odometry and PID control to achieve precise robot localization and path following.
- Designed and integrated a pneumatic system to improve game-piece manipulation and maximize scoring efficiency.
- Wired color and distance sensors to acquire feedback on an object's position for autonomous decision-making.
- Served as a certified VEX mentor, coaching middle school students in robot design, programming, and competition strategy.

Front-End Developer, LinkMe

June 2025 – July 2025

- Coded a full sign-up flow redesign using TypeScript and CSS.
- Added a new sign-up feature for business use that acquired 10 million new users for the app.
- Refined UI for better user experience, earning multiple positive reviews on the app store.

Technical Intern, ArtCube

March 2025 –

August 2025

- Collaborated with superiors during board meetings to brainstorm new product ideas.
- Trained an LLM to identify the tone of a video clip and suggest potential ads that could be placed.
- Coded algorithms in Python that manipulated videos (>30 minutes) to place sponsored products for clients.
- Satisfied investor needs and communicated deadlines, roadblocks, and challenges effectively.

Summer Camp Teacher, Code Ninjas

June 2024 –

August 2025

- Oversaw 10–20 students ages 7–14 weekly, teaching coding lessons in robotics, AI, and Minecraft.
- Helped set up games and projects used throughout the week.
- Explained challenging concepts to students, enabling them to complete tasks independently.
- Assisted with technology issues on classroom computers and communicated problems effectively.

Testing Intern, Venu AI

June 2024 –

August 2024

- Tested user cases for an AI model that automatically wrote and sent mass emails for business direct-marketing channels.
- Marketed the AI to external customers by creating presentations and hosting demos.
- Collaborated with peers in weekly meetings to resolve bugs and address customer complaints.

PROJECTS

Lead Engineer, Vision Assist SmartCane

May 2026 – Present

- Designed and developed an AI-powered smart cane that identifies nearby objects, estimates their distance, and provides real-time audio and tactile feedback for visually impaired users.
- Integrated an ESP32-CAM, ultrasonic distance sensors, vibration motors, Arduino-based electronics, and Bluetooth communication to enable obstacle detection and hazard alerts.
- Trained and deployed a TensorFlow computer vision model using Edge Impulse on a dataset of 16,000+ labeled images for real-time object recognition.
- Developed a companion website to deliver results through text-to-speech audio output.
- Modeled the cane in CAD and fabricated a working prototype.

Founder, FutureBuilders

December 2023 –

Present

- Founded a STEM nonprofit providing engineering and robotics education to 3,000 girls ages 8–21 in Libya.
- Wrote 50+ technical lessons on engineering design, mechanics, and FTC robotics for an online learning platform.
- Created and presented Java programming workshops demonstrating robot control, autonomous movement, and sensor integration.
- Collaborated with Lybotics to raise funding for hardware, computers, and learning materials for students.

Lead Engineer, TumbliTech

June 2025 – December 2025

- Designed and prototyped a compost tumbler to improve aeration and accelerate decomposition.
- Used CAD software to model components and create multiple design iterations.
- Engineered a reinforced chassis supporting 200+ lb of waste for maximum durability.
- Conducted performance testing and refined the design to improve usability.
- Donated the finished system to the school's Plant Club, which has composted 500+ lb of waste to date.

Independent Researcher, Lumiere

April 2025 –

October 2025

- Researched the effect of adversarial examples on Tesla's Full Self-Driving mode.
- Conducted hands-on research using a physical vehicle after obtaining IRB approval for testing.
- Paper reviewed and edited by NASA engineers, including a mentor and professor at the University of Michigan.
- Published in the Curieux Academic Journal.

SKILLS

Languages: C++, JavaScript, Java, Python, HTML/CSS, React/TypeScript

Embedded Systems: Arduino, ESP32-CAM, Bluetooth, Sensor Integration, Wiring, Soldering

Robotics / Control: PID Control, Odometry, TensorFlow, Edge Impulse

Software / Tools: Docker Desktop, Git/GitHub, Railway, Vercel, KiCad, Fusion 360, Blender

Mechanical: 3D Printing, Rapid Prototyping, Engineering Design